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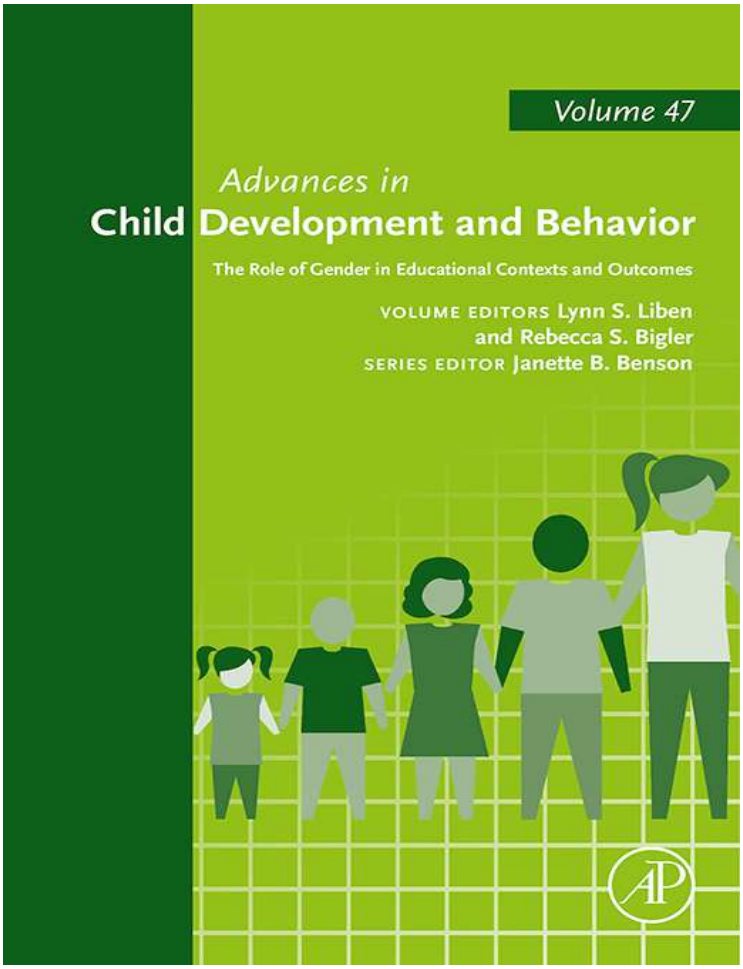

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From V. Paul Poteat, Jillian R. Scheer and Ethan H. Mereish, Factors Affecting Academic Achievement Among Sexual Minority and Gender-Variant Youth. In: Lynn S. Liben and Rebecca S. Bigler, editors, *Advances in Child Development and Behavior, Vol. 47*, Burlington: Academic Press, 2014, pp. 261-300.
ISBN: 978-0-12-411582-8
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CHAPTER EIGHT

Factors Affecting Academic Achievement Among Sexual Minority and Gender-Variant Youth

V. Paul Poteat[1], Jillian R. Scheer, Ethan H. Mereish
Department of Counseling, Developmental, and Educational Psychology, Boston College, Chestnut Hill, Massachusetts, USA
[1]Corresponding author: e-mail address: PoteatP@bc.edu

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The Role of Gender in Educational Contexts and Outcomes (L. S. Liben & R. S. Bigler, Eds.)
Advances in Child Development and Behavior (J. B. Benson, Series Ed.), Vol. 47
ISSN 0065-2407 http://dx.doi.org/10.1016/bs.acdb.2014.04.005
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Abstract

Experiences of victimization among sexual minority youth (e.g., lesbian, gay, bisexual, transgender; LGBT) and gender-variant youth remain pronounced in many schools. Although much work has shown the connection between homophobic bullying and mental and physical health, there has been limited attention to how victimization impedes learning, academic achievement, and other school-related outcomes for these youth. In this chapter, we propose several pathways through which victimization leads to academic disparities among sexual minority and gender-variant youth, with attention to its effects on individual learning processes (e.g., motivation, concentration, self-efficacy, and other cognitive stressors) as well as broader psychological and social processes (e.g., mental health, school avoidance, harmful coping strategies, exclusionary discipline). We also consider protective factors (e.g., social support, Gay–Straight Alliances, extracurricular involvement, nondiscrimination policies, inclusive curriculum) that could promote resilience and suggest potential mechanisms by which they may operate. In doing so, we aim to stimulate ideas for an advancement of research in this area.

1. INTRODUCTION

Schools are a critical setting for many aspects of youth development, ranging from academic to social development (Eccles & Roeser, 2011; Larson, Hansen, & Moneta, 2006). Nevertheless, many youth continue to experience hostility, rejection, and victimization in this context and perceive their schools as unsafe and unwelcoming (Cook, Williams, Guerra, Kim, & Sadek, 2010). These experiences carry serious physical and mental health consequences, impede learning processes, and act as barriers to academic achievement for these youth (Gini & Pozzoli, 2009; Nakamoto & Schwartz, 2010; Reijntjes, Kamphuis, Prinzie, & Telch, 2010; van Lier et al., 2012).

Experiences of victimization among sexual minority youth (e.g., lesbian, gay, bisexual, transgender; LGBT) remain particularly pronounced in many schools. In fact, several recent meta-analyses show that sexual minority youth experience higher rates of school-based victimization than heterosexual youth and that this constitutes a sizable difference (Katz-Wise & Hyde, 2012; Toomey & Russell, 2013b). Further, there is robust support for strong associations between homophobic victimization, discrimination, and a range of mental health, physical health, and academic concerns (Friedman, Koeske, Silvestre, Korr, & Sites, 2006; Hershberger & D'Augelli, 1995; Huebner, Rebchook, & Kegeles, 2004; Poteat, Mereish, DiGiovanni, & Koenig, 2011; Russell, Sinclair, Poteat, & Koenig, 2012).

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Similarly, youth experience victimization on the basis of their gender identity or expression, particularly for behaviors that violate rigidly prescribed masculinity or femininity norms (Aspenlieder, Buchanan, McDougall, & Sippola, 2009; D'Augelli, Grossman, & Starks, 2006; Pascoe, 2007; Phoenix, Frosh, & Pattman, 2003; Toomey, McGuire, & Russell, 2012; Toomey, Ryan, Diaz, Card, & Russell, 2010). It is important to note that sexual orientation and gender identity and expression are not synonymous. Both sexual minority and heterosexual youth vary in the extent to which they express behaviors traditionally labeled as masculine or feminine, and thus both may experience victimization when they violate gender norms (Friedman et al., 2006; Pascoe, 2007; Phoenix et al., 2003). At the same time, due to their shared experiences of discrimination, transgender youth often have been grouped with lesbian, gay, and bisexual youth when researchers make broader comparisons between sexual minorities and heterosexuals. Nevertheless, it is important to note that transgender youth (as well as gender-variant youth more broadly) may identify as heterosexual or as lesbian, gay, or bisexual. Because most of the research we review in this chapter has not distinguished between transgender youth who identify as lesbian, gay, or bisexual, and those who identify as heterosexual, for the purpose of this chapter we refer broadly to the sexual minority youth community to include transgender youth. As with victimization based on sexual orientation, victimization based on gender expression is associated with elevated health and academic concerns (Grossman & D'Augelli, 2006; Toomey et al., 2010). Thus, both forms of bias-based victimization affect school and academic outcomes.

Research on sexual minority and gender-variant youth has focused heavily on their experiences of victimization and other stressors specifically within the school context. Taking into consideration this focus on school-based experiences, it is surprising that there has been rather limited direct attention to various academic and career development concerns faced by many of these youth. Nevertheless, some extant findings do indicate sexual orientation-based academic disparities in several areas such as enrollment in advanced courses, earned grades, and risk of school dropout (Aragon, Poteat, & Espelage, 2014; Pearson, Muller, & Wilkinson, 2007; Poteat et al., 2011; Russell, Seif, & Truong, 2001). Undeniably, continued attention to mental and physical health disparities is critical. A number of comprehensive reviews on the health disparities faced by sexual minorities emphasize the continued need for this research (e.g., Graham, Bradford, de Vries, & Garofalo, 2011). At the same time, it is important for researchers to extend their focus on sexual minority and gender-variant youth to include

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issues related to learning, academic performance, and career development. These educational and career issues are fundamentally interconnected with mental and physical health issues. It is clear from the general psychological literature, for example, that mental health concerns predict academic and other school-related concerns; furthermore, the effects of victimization on academic outcomes are partly mediated through their negative effects on mental health (DeRosier & Mercer, 2009; Juvonen, Nishina, & Graham, 2000; Kochenderfer & Ladd, 1996). Given the limited attention to academic issues among sexual minority and gender-variant youth, we seek to highlight current issues in this area and suggest new avenues for research.

We have several primary aims for this chapter. First, we bridge concepts from multiple theories, including minority stress theory and theories of learning and career development, to provide a framework on which to base research on academic disparities faced by sexual minority and gender-variant youth. Second, we synthesize findings on the existence of sexual orientation-based academic disparities. Third, we consider the role of victimization and other stressors in shaping the school-based experiences of sexual minority and gender-variant youth. We offer several suggestions for how these barriers may operate in combination to predict school and academic outcomes among sexual minority and gender-variant youth. Although we emphasize the role of victimization and the school context, we acknowledge that victimization is only one of many stressors experienced by sexual minority and gender-variant youth and that school is one of multiple contexts in which it is experienced. Fourth, we consider factors that moderate and attenuate the effects of victimization and other disruptive processes that impede learning and academic performance. Finally, we discuss relevant programming and policy issues at a broader systems level. As part of this discussion, we suggest several pathways by which they may improve the school-based experiences and academic achievement of sexual minority and gender-variant youth.

2. THEORETICAL MODELS FOR UNDERSTANDING ACADEMIC DISPARITIES

In this section, we review two major theories that we apply throughout this chapter as frameworks for understanding academic disparities faced by sexual minority and gender-variant youth. Both models suggest various pathways through which disparities may arise for these youth. The minority

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stress model (Meyer, 2003) has been widely used to explain sexual orientation-based disparities related to mental and physical health, but it has been used markedly less when considering academic disparities. In comparison, social cognitive career theory (SCCT; Lent, Brown, & Hackett, 1994, 2000) focuses directly on individual and social processes that influence learning and career development. SCCT has been applied in some limited cases to identify barriers to career development among sexual minority adults, but with little attention to sexual minority youth. In combination, however, these two theories provide a strong framework to identify processes that could contribute to sexual orientation-based academic disparities.

2.1. The Sexual Minority Stress Model

The minority stress model (Meyer, 2003) offers one framework to understand sexual orientation-based disparities in school and academic outcomes. The minority stress model posits that sexual minorities face unique and hostile stressors (e.g., homophobic victimization) related to their sexual minority identity; consequently, these stressors have negative effects on their health (Meyer, 2003). Although the model was originally conceptualized for sexual minorities, it can be applied similarly to gender-variant individuals. The model outlines distal and proximal stressors that are unique and chronic for minority populations. Distal stressors may constitute experiences such as discrimination, while proximal stressors represent more internally based processes such as internalized homonegativity (Meyer, 2003). Although the model has received much empirical support, its focus has been on mental and physical health outcomes with little attention to academic outcomes. We believe, however, that the model offers a useful lens to conceptualize sexual minority and gender-variant youths' school experiences and academic outcomes.

2.2. Social Cognitive Career Theory

As a complement to the minority stress model, we utilize SCCT (Lent et al., 1994, 2000) to frame our discussion of sexual minority and gender-variant youths' experiences in the school and academic domain. Although SCCT focuses on career development, we consider educational and academic experiences to be part of and contributing to this long-term process. Of note, much of SCCT is grounded in Bandura's work on social learning theory and self-efficacy (Bandura, 1977).

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SCCT proposes that individuals' personal attributes (e.g., sexual orientation) and their context (e.g., environmental conditions and events) influence their learning experiences (Lent et al., 1994, 2000). For instance, and as we later note, contextual factors such as unwelcoming school climates and victimization can exert a sizeable negative effect on the learning experiences of sexual minority and gender-variant youth. In turn, these learning experiences influence individuals' self-efficacy, interests, outcome expectations, and personal goals. Self-efficacy refers to confidence in one's abilities (e.g., to perform well in a course or graduate from high school). Self-efficacy can be influenced by personal accomplishments (e.g., prior academic performance), vicarious learning (e.g., observing teachers model ways to solve a problem), persuasion (e.g., receiving positive reinforcement from teachers), and physiological states (e.g., physical health). Self-efficacy is important because it can summarily influence individuals' continued interest in an area (e.g., continued interest in science, sports, music). Both self-efficacy and interest are related to individuals' outcome expectations (i.e., beliefs about what will result from one's behavior and performance). Outcome expectations, in turn, shape individuals' personal goals and aspirations. Butler (2014) [Chapter 1 of this volume] and Petersen and Hyde (2014) [Chapter 2 of this volume] provide accounts of these processes in the development of gender-differentiated academic motivation and achievement. As an example related to sexual minority youth, a lesbian adolescent's intent to remain in high school may be shaped by her expectations of continued victimization, level of encouragement from teachers to stay in school, and from her past academic success.

In addition to the set of factors above, SCCT emphasizes that personal and contextual factors can affect learning and career development (Lent et al., 2000). These components are critical, considering the unique barriers faced by sexual minority and gender-variant youth that disrupt their learning and impede their academic performance. For instance, distal and proximal stressors outlined by the minority stress model may act as barriers and negatively affect the learning process outlined by SCCT. One study among sexual minority college students, for example, found that discrimination was associated negatively with career development (Schneider & Dimito, 2010). Ultimately, we suspect that chronic victimization experienced by sexual minority and gender-variant youth may influence academic performance through its negative effects on self-efficacy, outcome expectations, and other related pathways.

Overall, sexual minority and gender-variant youth face unique barriers to achieving their academic and career goals. The minority stress model and SCCT provide useful frameworks to consider the complex processes that lead

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to academic disparities. In the sections that follow, we apply these frameworks to highlight potential pathways that should be tested empirically in order to address the academic needs of sexual minority and gender-variant youth.

3. EVIDENCE OF SEXUAL ORIENTATION-BASED ACADEMIC DISPARITIES

There is a sizable literature base on the achievement gap and educational disparities faced by minority youth populations, particularly racial and ethnic minority youth (Cohen, Garcia, Apfel, & Master, 2006; Ladson-Billings, 2006; Lee, 2002; Rothstein, 2004). Comprehensive reviews have highlighted issues such as inequitable and limited access to educational resources, the discrepant quality of education provided, concerns related to academic assessment and testing, and differential treatment by educators that contribute to race- and ethnicity-based academic disparities. Nevertheless, studies have systematically overlooked and failed to consider the academic needs of sexual minority youth and potential sexual orientation-based academic disparities. Furthermore, educational policies have failed to address issues of school accountability to ensure the academic success of these youth. In this section, we review the limited findings on school and academic concerns among sexual minority youth, as well as concerns related to their future academic intentions and career aspirations.

Basic disparities between sexual minority and heterosexual youth have been documented across multiple school-related and academic indices. School safety and belonging have been among the most extensively examined of these indices. Many sexual minority youth perceive their schools as unsafe and unwelcoming, and their perceptions of safety and belonging are lower compared to heterosexual youth (Bochenek & Brown, 2001; Eisenberg & Resnick, 2006; Murdock & Bolch, 2005; Poteat et al., 2011; Toomey et al., 2012). Several crucial components of school safety and belonging have been considered as part of this research, such as the availability of a supportive adult, feelings of isolation or disconnection, attachment to teachers or peers, and perceptions of the school as a welcoming environment. As we later note, school belonging is especially important to consider based on its association with victimization and academic outcomes.

Other studies have given attention to disparities related to course grades and GPA. For example, some findings have noted that, on average, sexual minority youth report poorer grades than heterosexual youth (Poteat et al., 2011). When considering gender differences, however, other findings

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suggest that this applies primarily to gay and bisexual males rather than to lesbian or bisexual females (Pearson et al., 2007). Still other findings have indicated that GPA differences are more pronounced for male and female bisexual youth than they are for same-sex-only attracted youth (Russell et al., 2001). Thus, although grade disparities appear evident, there remains ambiguity as to which groups of sexual minority youth they most apply.

Studies also have considered differences between heterosexual and sexual minority youth in the courses they take in high school. Some findings have shown that sexual minority males are less likely to take advanced math and science courses than heterosexual males, and that sexual minority females are less likely to take advanced science courses than heterosexual females (Pearson et al., 2007). These course-related differences may explain why some sexual minority youth maintain a relatively strong GPA, despite their frequent victimization. These findings are concerning for several reasons. Advanced courses are critical for competitive college acceptance and as part of preparation for the rigor of college-level courses. As such, even though sexual minority youth may perform well in their classes, taking fewer rigorous classes places them at a competitive disadvantage for longer-term academic opportunities. Disadvantages may be evident in areas such as admittance to highly competitive universities, securing academic merit scholarships, or fulfilling prerequisite courses for certain college majors. Research is needed to consider these potential disparities.

These course-related disparities also may carry implications for longer-term career development. There continues to be substantial growth and career opportunities in STEM-related fields (i.e., science, technology, engineering, and mathematics). Following the SCCT model, however, sexual minority youth who take fewer courses in these areas may feel lower self-efficacy to perform well in these areas; subsequently, their lower self-efficacy may lead them to express less interest in pursuing STEM-related careers. Of importance, we suspect that sexual orientation-based differences in course enrollment are not simply a reflection of different interests between sexual minority and heterosexual youth. We later note how victimization may operate as a barrier to taking and excelling in advanced courses. These emerging findings underscore the need for research to consider not only immediate sexual orientation-based academic disparities, but also whether these disparities extend beyond secondary education and are compounded over time.

Adding to the concerns noted above, heterosexual and sexual minority youth differ in their projected future academic intentions. One study found

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that sexual minority youth were more than twice as likely as heterosexual youth to report that they would not complete high school, and they were significantly less likely to report intentions to go to a 4-year college (Aragon et al., 2014). There is little disagreement that a college degree is now a minimal standard for many careers. Thus, sexual minority youth who do not complete high school or who do not pursue a full college degree may face more limited career options after high school.

As research continues to advance in this area, researchers also must give greater consideration to diversity within the sexual minority youth community. Academic disparities may be even more evident when considering the intersection of sexual orientation with other social identities. Whereas some studies have considered gender differences (e.g., Pearson et al., 2007; Russell et al., 2001), other forms of diversity also should be included. Given the notable education disparities faced by youth of color (Barbarin, Chinn, & Wright, 2014 [Chapter 10 of this volume]; Ladson-Billings, 2006; Lee, 2002; Rowley et al., 2014 [Chapter 9 of this volume]), combined with findings that sexual minorities of color can experience unique stressors from White sexual minorities (Meyer, Dietrich, & Schwartz, 2008; Rosario, Schrimshaw, Hunter, & Gwadz, 2002), academic disparities may be especially pronounced for sexual minority youth of color. Similarly, there is a dearth of research on transgender youth in relation to academic outcomes. Yet, they face many of the same barriers as LGB youth, as well as other unique barriers (Grossman & D'Augelli, 2006; Mallon & DeCrescenzo, 2006; McGuire, Anderson, Toomey, & Russell, 2010). Attention to variability within the sexual minority youth community is important as part of larger efforts to ensure that adequate resources are provided to these youth and that these resources are tailored to meet the needs of specific populations within this community.

4. PROCESSES AND CONSEQUENCES OF VICTIMIZATION

4.1. Homophobic Victimization and Academic Disparities

Among the distal stressors outlined in minority stress theory (Meyer, 2003), homophobic victimization represents a prominent one faced by sexual minority youth. Name-calling, rumor-spreading, teasing, and assault are among the most common forms of victimization experienced by these youth (Kosciw, Diaz, & Greytak, 2008; Rivers, 2001). Of concern, findings from multiple data sources over several decades show little evidence of a decrease in the large proportion of sexual minority youth who face this form of

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victimization (Hershberger & D'Augelli, 1995; Kosciw, Greytak, & Diaz, 2009; Rivers, 2001). A few longitudinal and retrospective findings indicate that many sexual minority youth experience chronic and prolonged victimization (Rivers, 2001; Rosario et al., 2002). For example, sexual minority adults have reported retrospectively that their victimization at school often was ongoing (Rivers, 2001). Prospective longitudinal studies also indicate this relative stability. Sexual minority youth who experience higher levels of gay-related stress than other youth continue to report higher levels than other youth over a 12-month period (Rosario et al., 2002). As we will note, such persistent victimization stands to have a substantial negative effect on academic performance.

Our knowledge of victimization based on gender expression has grown with recent research. Gender-variant behavior is associated with victimization for heterosexual and sexual minority youth (Aspenlieder et al., 2009; D'Augelli et al., 2006; Pascoe, 2007; Young & Sweeting, 2004). Other studies among sexual minority youth note the association between victimization based on gender expression and mental health concerns (Friedman et al., 2006; Toomey et al., 2012). Based on retrospective reports from sexual minority adults, higher levels of recalled victimization based on gender expression during adolescence predicted their current life satisfaction (Toomey et al., 2012) and adult gay men's current suicidality (Friedman et al., 2006). There remains little attention, however, to academic disparities faced by gender-variant youth.

Several meta-analyses provide a characterization of the victimization faced by sexual minority youth. One meta-analysis found that sexual minority youth reported higher levels of multiple forms of victimization than heterosexual youth, with small to moderate effect sizes (Katz-Wise & Hyde, 2012). Some of these differences were slightly higher for males than females. Another meta-analysis specific to school-based victimization came to similar conclusions and deemed the overall magnitude of sexual orientation-based differences to be moderate (Toomey & Russell, 2013b). Furthermore, there was some evidence that the disparity was greater for males than females.

Some scholars have argued that sexual prejudice and discrimination have dissipated among youth and that the serious health risks faced by sexual minority youth have been exaggerated or no longer apply to many of these youth (McCormack, 2012; Savin-Williams, 2005). The empirical evidence, however, is largely unsupportive of or runs counter to these claims. At best, the prevailing evidence suggests that this counter-argument applies to a small subsample of sexual minority youth. This is not to diminish the case that

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many sexual minority youth are resilient in the face of discrimination and that others demonstrate healthy development in physical, psychological, social, and academic domains. Nevertheless, a large number of sexual minority youth continue to experience chronic victimization and related hardships in each of these domains. As such, the nature and prevalence of these concerns cannot be minimized.

Although studies offer firm evidence that sexual minority youth report higher rates of victimization than heterosexual youth, it is important to note that heterosexual youth also are called homophobic epithets and experience victimization that is homophobic (Pascoe, 2007; Phoenix et al., 2003; Plummer, 2001; Poteat & Espelage, 2007; Poteat, Scheer, DiGiovanni, & Mereish, 2014). Notably, homophobic victimization is associated with psychosocial and academic concerns for heterosexual youth in a similar manner as for sexual minority youth (Poteat & Espelage, 2007; Poteat, Scheer, et al., 2014). Homophobic victimization among heterosexual youth has been explained based on several factors. Because of the societal stigmatization of sexual minorities, referring to a student as a sexual minority in a disparaging manner, regardless of whether the student identifies as such, symbolically places that student in a subordinate position. Also, using homophobic epithets is one means by which aggressive students intensify bullying behavior, irrespective of the targeted individual's actual or perceived sexual orientation (Plummer, 2001; Thurlow, 2001). Finally, these epithets are used to enforce gender-normative behavior and punish gender-transgressive behavior, particularly among boys (Pascoe, 2007; Phoenix et al., 2003; Plummer, 2001). This latter dynamic highlights some of the overlap between homophobic victimization and victimization based on gender-variant behavior.

Congruent with assertions that homophobic bias intensifies aggressive acts, the associations between homophobic victimization and academic concerns are greater in size than for general victimization (e.g., victimization that students do not perceive to be based on their sexual orientation). Specific to school and academic outcomes, Russell and colleagues (2012) found that students who experienced general victimization were 1.32 times as likely as nonvictimized youth to report that their grades were mostly C's or below and they were 1.79 times as likely to report truancy. This disparity was greater for youth who had experienced homophobic victimization: they were 1.52 times as likely as nonvictimized youth to report that their grades were mostly C's or below and 2.52 times as likely to report truancy. This study was unable to differentiate heterosexual and sexual minority youth, yet past research suggests that sexual minority youth experience more

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homophobic victimization than heterosexual youth (Katz-Wise & Hyde, 2012; Poteat et al., 2011; Toomey & Russell, 2013b). Homophobic victimization may result in these heightened detrimental effects for sexual minority youth because it directly denigrates their marginalized social identity. Consequently, it may not simply be that sexual minority youth experience more frequent victimization than heterosexual youth, but that the victimization they experience (i.e., homophobic victimization) is also more severe.

Chronic victimization among sexual minority youth may partly explain how sexual orientation-based academic disparities arise. Although short-term stressors (e.g., a brief illness) may impede learning or affect academic performance temporarily, students may be able to improve without a lasting consequence. Chronic victimization, however, jeopardizes students' learning and performance across multiple assignments and tests. Moreover, because later educational material often builds on earlier material, the effects of chronic homophobic victimization on academic performance may be compounded over time for sexual minority youth. It may be much more difficult for these youth to recover academically, which ultimately may affect their final grades. Over an even greater time period, this may limit their opportunities to enroll in advanced courses. Ultimately, chronic victimization could partly account for findings that sexual minority youth enroll in fewer advanced math or science courses than heterosexual youth (Pearson et al., 2007), a possibility in need of empirical examination.

4.2. Processes by Which Victimization Affects Learning and Academic Performance

Victimization operates in combination with other factors in ways that lead to poorer educational outcomes among sexual minority than heterosexual youth. Although most of this research has been conducted among sexual minority youth, we believe it is also relevant for gender-variant youth (either sexual minority or heterosexual). In this section, we review the connections between victimization and other psychological and social factors that shape the experiences of these youth. In addition, we point to several potential mechanisms by which these factors could lead to poor learning and academic outcomes.

4.3. Diminished Mental and Physical Health

A number of studies have focused on the mediating role of mental health in accounting for the association between victimization and academic outcomes. Specifically, victimization predicts diminished mental health, which in turn

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predicts elevated school and academic concerns such as truancy, poorer grades, and risk for school dropout (Juvonen et al., 2000). There is evidence that this also applies to sexual minority youth (Poteat et al., 2011).

There are several potential reasons for why mental health concerns resulting from victimization have a central role in predicting academic outcomes for sexual minority and gender-variant youth. In addition to the direct disruptive effect of victimization on learning (Boulton, Trueman, & Murray, 2008), mental health concerns such as depression can further impede learning processes. These learning processes can include concentration, motivation to learn the material, active participation and engagement in the learning process, and the ability to retain and recall information (Buhs & Ladd, 2001). Mental health concerns also may lead youth to miss days from school (Egger, Costello, & Angold, 2003), which further disrupts the learning process. Given that sexual minority and gender-variant youth experience more frequent victimization than heterosexual and gender-typical youth (Katz-Wise & Hyde, 2012), that homophobic victimization is more severe than general victimization (Russell et al., 2012), and that many sexual minority and gender-variant youth have fewer support outlets than heterosexual youth (Goodenow, Szalacha, & Westheimer, 2006; Ryan, Huebner, Diaz, & Sanchez, 2009), it is perhaps unsurprising that mental health concerns have a major role in accounting for academic concerns among sexual minority and gender-variant youth.

In a similar manner, physical health concerns also may impede learning and academic performance among sexual minority and gender-variant youth. There is robust support for the association between victimization and physical health concerns (Gini & Pozzoli, 2009; Nishina, Juvonen, & Witkow, 2005), and there is additional evidence that physical health concerns are even higher among sexual minorities than heterosexuals (Cochran, Mays, Alegria, Ortega, & Takeuchi, 2007; Sandfort, Bakker, Schellevis, & Vanwesenbeeck, 2006). Physical health impairments may disrupt learning through its effects on impaired concentration or forcing students to miss classes (e.g., for doctor appointments, pain management, complications arising from the primary health concern). These findings highlight the need to consider not only mental health concerns but also physical health concerns that contribute to sexual orientation-based academic disparities.

4.4. School Avoidance

Certain protective strategies used by sexual minority and gender-variant youth to avoid victimization also carry risks connected to their academic performance. For instance, some sexual minority youth report school

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avoidance and absenteeism out of serious safety concerns (Goodenow et al., 2006; Poteat et al., 2011). Often, these youth must contend with immediate safety needs that supersede their ability to focus on academic tasks and goals. Several factors contribute to sexual minority and gender-variant youth engaging in school avoidance as a protective strategy. A number of sexual minority youth report that teachers often fail to intervene when students make homophobic comments or use homophobic epithets (Elze, 2003). Similarly, documents from lawsuits filed by sexual minority and gender-variant youth against their school systems have shown that these youth often were blamed for their victimization, were at times viewed as instigators of these experiences, and faced added discrimination from adults and authority figures at school (Cianciotto & Cahill, 2012). In the absence of support or protection from adults at school, sexual minority and gender-variant youth may view school avoidance as one of few ways to protect themselves, despite its consequences in other areas (e.g., learning and academic performance).

School avoidance may negatively affect academic outcomes through several pathways. Repeated truancy presents a barrier to sexual minority and gender-variant students' access to course material and participation in class-based learning activities. Also, it isolates them from other teachers and peers who may be in a position to provide academic support. In relation to this latter issue, help avoidance may be a particular concern for sexual minority and gender-variant youth. Help avoidance is a significant barrier to learning and academic achievement in general (Butler, 1998; Ryan, Patrick, & Shim, 2005). For sexual minority and gender-variant youth in particular, some may avoid help seeking because they anticipate being treated in a biased manner or fear rejection from their teachers on the basis of their sexual orientation or gender expression. Some youth also may hesitate to access help from teachers who have failed to intervene during instances of homophobic harassment (Elze, 2003). This underscores the need to work with teachers on ways to serve as allies for sexual minority and gender-variant youth, and to foster stronger teacher-student connections with these youth in ways that could promote both their well-being and academic achievement.

4.5. Substance Use as an Externalizing Coping Strategy

Some sexual minority and gender-variant youth use other strategies to cope with victimization that place their academic performance at risk. Studies have documented the connection between victimization and substance

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use among sexual minority youth (Bontempo & D'Augelli, 2002; Marshal et al., 2008). Some sexual minority youth use substances to cope with chronic victimization because other support structures are not accessible (Rosario, Schrimshaw, & Hunter, 2011). Nevertheless, this form of coping carries negative implications for school and academic outcomes. As with diminished mental and physical health, substance use can inhibit learning processes such as concentration, motivation, and information retention and recall (Bryant, Schulenberg, O'Malley, Bachman, & Johnston, 2003). Thus, although some sexual minority and gender-variant youth may experience internalizing concerns (e.g., depression or anxiety) and others may experience externalizing concerns (e.g., truancy or substance use), these two types of coping responses may operate through similar pathways to undermine academic achievement.

4.6. Chronic Vigilance

The minority stress model (Meyer, 2003) suggests several other internal processes by which homophobic victimization may disrupt the learning and academic performance of sexual minority and gender-variant youth. As noted in the model, distal stressors (e.g., discrimination) can contribute to other proximal stress processes. One such stressor is heightened and chronic vigilance to guard against potential discrimination and rejection (Meyer, 2003). Vigilance is an especially relevant factor when considering issues related to learning and academic performance. Heightened vigilance to ensure physical safety likely comes at the expense of time and energy available to engage in other actions (e.g., studying) that serve to promote higher-order academic needs.

The effects of vigilance on academic achievement may become more pronounced as sexual minority and gender-variant youth progress through school. As students take more advanced classes, learning the material requires a substantially greater cognitive load (e.g., it requires a higher level of abstract reasoning, problem-solving, simultaneous consideration of multiple pieces of information; Sweller, 1988; van Merriënboer & Sweller, 2005). Thus, students must invest a much greater amount of working memory to attend to, process, retain, and eventually reproduce the information taught in these courses. Many sexual minority and gender-variant youth must contend with chronic vigilance in the face of perpetual discrimination. This additional cognitive demand is likely to reduce their capacity to master the material in these courses.

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There has been virtually no attention to the role of vigilance in relation to learning and academic performance among sexual minority and gender-variant youth. This factor could have a significant role in explaining academic concerns among these youth, over and above effects tied to internalizing or externalizing concerns. Indeed, this process could partly explain the basic finding that sexual minority youth tend to enroll in less advanced math and science courses than their heterosexual peers (Pearson et al., 2007). Sexual minority and gender-variant youth may enroll in less demanding courses based on their anticipation of poor performance due to these added distal (e.g., homophobic victimization) and proximal (e.g., chronic vigilance) stressors.

5. ADDITIONAL INFLUENCES ON STUDENT OUTCOMES

5.1. The Timing and Onset of Victimization as a Risk Factor

The time period during which sexual minority and gender-variant youth begin to experience victimization is important to consider. Heterosexual and sexual minority youth have reported that homophobic victimization often began in elementary school (Kosciw et al., 2008; Plummer, 2001). Findings also suggest that the school climate for sexual minority youth is especially hostile during middle school compared to high school. In one large study of heterosexual youth in grades 7–12, over 40% of students in grade 7 strongly agreed or agreed with the statement that they could "never stay friends with someone who told me that he or she was gay or lesbian," while 45% of students in grade 7 strongly agreed or agreed with the statement that they would "rather attend a school where there are no gay or lesbian students" (Poteat, Espelage, & Koenig, 2009). The proportion of students who endorsed these items steadily decreased over corresponding grade levels. Nevertheless, students' support for such blatant rejection of sexual minority youth presents a stark representation of the climate in which many sexual minority youth must attempt to learn at such formative developmental periods.

These patterns of victimization and rejection should be considered as they overlap with the coming out process for sexual minority youth. Younger generations of sexual minorities are coming out at earlier ages than older generations (Floyd & Bakeman, 2006; Grov, Bimbi, Nanín, & Parsons, 2006). Some findings have placed the average age of coming out among contemporary sexual minority youth as between 14 and 15 or between 16 and 17 years of age (Floyd & Bakeman, 2006; Grov et al., 2006). These averages are distinct from those of older generations

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of sexual minorities, whose general age range of coming out was from 24 to 27 years of age (Grov et al., 2006). In other words, whereas older generations of sexual minorities tended to come out well after completing their secondary education (or even higher education for those who were able to attend college), contemporary youth now come out, on average, during high school. As an average, this suggests that other sexual minority youth come out even earlier. This represents a critical issue for psychologists and educators when these trends are paired with findings that prejudice against sexual minorities is at its highest among younger age groups (Horn, 2006; Poteat & Anderson, 2012). As sexual minority youth begin to come out at earlier ages, they also face a heightened potential for peer rejection. This underscores the need for schools to ensure the safety of these youth, to provide them with supportive resources, and to address the broader school climate.

These co-occurring processes (i.e., emergence of homophobic victimization, developmental trajectories of prejudice, and the coming out process) intersect with educational processes and academic expectations at these same points in time. Discrimination and hostility toward sexual minority youth peak at a time when academic expectations are becoming more demanding (e.g., during the transition from middle school to high school) and when students are being considered for placement in advanced courses. Thus, although Pearson and colleagues (2007) found that sexual minority youth were less likely than heterosexual youth to have taken advanced math and science courses in high school, it seems possible that sexual orientation-based academic disparities emerge earlier than high school. If so, they may partly result from the co-occurring stressors noted above that sexual minority and gender-variant youth contend with in their early school-based experiences. The immediate effects of early victimization on academic performance may lead to prolonged academic consequences for sexual minority and gender-variant youth. Again applying the SCCT model (Lent et al., 1994), poorer grades could lead these youth to face restricted course options in the future, lower their academic self-efficacy, and lead them to foreclose on pursuing their interests in certain careers. To our knowledge, longitudinal research has not yet addressed these issues. It is critical for research to test whether sexual orientation-based academic disparities in high school are a product not only of immediate circumstances (e.g., current victimization) but also of a prolonged history of discrimination and an extensive sequence of negative academic outcomes. This information would carry significant implications for interventions to counter these disparities.

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5.2. Exclusionary Discipline as a Disruptive Process

There has been a recent emergence of attention to exclusionary discipline and its effects among sexual minority and gender-variant youth. Although it has been studied only recently among this youth population, exclusionary discipline has been studied extensively among other minority youth populations. Exclusionary forms of discipline, ranging from school office referrals, to suspensions, to incarceration, are directed disproportionately toward racial minority youth (for reviews, see Gregory, Skiba, & Noguera, 2010; Wallace, Goodkind, Wallace, & Bachman, 2008). Furthermore, these youth are more likely to face harsher sanctions than White youth who report similar infractions (Piquero, 2008; Skiba, Michael, Nardo, & Peterson, 2002). Based on the empirical data, researchers have argued persuasively that there may be a strong connection between discipline bias and the achievement gap among racial minority youth (Gregory et al., 2010). These forms of exclusionary discipline can be disruptive to the learning process and may be linked to a number of academic risks such as school dropout (Ekstrom, Goertz, Pollack, & Rock, 1986). Causes, consequences, and solutions to the problem of racial biases in the use of exclusionary discipline are addressed by Barbarin et al. (2014) [Chapter 10 of this volume] and Rowley et al. (2014) [Chapter 9 of this volume].

Building on this established literature base, there is initial evidence that sexual minority youth experience exclusionary discipline at higher rates than heterosexual youth. Himmelstein and Brückner (2011) found that non-heterosexual youth were more likely than heterosexual youth to report juvenile arrests and convictions. In addition, we have found that sexual minority youth were more likely to have been suspended from school and to have been involved in the juvenile justice or prison system than heterosexual youth (Poteat, Scheer, & Chong, under review). These experiences have the potential to act as significant barriers to learning and academic performance for sexual minority youth.

It is important to identify factors that contribute to the existence of these discipline disparities. In our recent work (Poteat et al., under review), we used the minority stress model (Meyer, 2003) as a framework to demonstrate that victimization is again a key factor that contributes to these disparities. Specifically, we built on the well-established connection between victimization and externalizing behaviors (e.g., substance use, truancy; Goodenow et al., 2006; Rosario et al., 2011) to show that these factors predicted increased risk for suspension and juvenile justice system involvement. This extended link may have been evident because, as minors, some of the very

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ways in which sexual minority youth cope with discrimination are punishable infractions. Moreover, we found evidence suggestive of bias against sexual minority youth in the disciplinary process: these externalizing behaviors were stronger predictors of suspension and juvenile justice system involvement for sexual minority youth than heterosexual youth.

To our knowledge, research has not yet examined how exclusionary discipline leads to poorer academic outcomes among sexual minority and gender-variant youth. We suspect, however, that exclusionary discipline creates significant academic hardships for these youth. Indeed, the model we have proposed is itself a cascade of stressors experienced by these youth, extended to include exclusionary discipline. In this sense, many sexual minority and gender-variant youth who experience exclusionary discipline also are coping with other stressors that place them at risk for academic concerns. Furthermore, sexual minority and gender-variant youth who are temporarily restricted from attending school may be even less likely to return (i.e., more likely to drop out) compared to heterosexual and gender-typical youth. This possibility should be explored, as there are several circumstances that could contribute to such an increased risk. Sexual minority youth, more than heterosexual youth, already may perceive their school to be a hostile and unsupportive environment in which they feel disconnected from adults and peers. In short, there is a pressing need for research and intervention work to address the issue of exclusionary discipline among sexual minority youth while also attending to the underlying causes (e.g., mental health concerns, safety concerns) that place these youth at greater risk.

5.3. Unique Stressors Among Transgender Youth

Up to this point, we have focused our discussion on sexual minority youth, broadly considered to include lesbian, gay, bisexual, and transgender youth. However, most of the research on sexual minorities either has not included transgender youth participants or these youth were significantly underrepresented. Nevertheless, transgender youth experience many of the same stressors we have reviewed earlier. Some transgender youth are victimized at school because they are perceived to be gay or lesbian and they may also experience victimization for their gender expression or violation of rigid gender norms (Grossman & D'Augelli, 2006; Mallon & DeCrescenzo, 2006; McGuire et al., 2010). There is some evidence that suggests transgender youth face even greater victimization than their lesbian, gay, and bisexual peers (McGuire et al., 2010). Like other sexual minority youth, transgender youth report heightened

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health and academic risks as a consequence of victimization (e.g., depression, suicidality, truancy, lack of safety, risk of school dropout; Garofalo, Deleon, Osmer, Doll, & Harper, 2006; McGuire et al., 2010; Rosenberg, 2002). In effect, these discriminatory experiences could impede learning and academic performance for transgender youth in the same way as for other sexual minority youth.

At the same time, it is important to highlight several unique stressors faced by transgender youth within the school context and the implications of these stressors for learning and academic performance. Some of these additional stressors include school resistance to referring to transgender youth by their affirmed gender (i.e., the gender with which they personally identify) and preventing them from using the bathroom and locker room of their affirmed gender or playing on the sports team of their affirmed gender (Barron & Bradford, 2007; McGuire et al., 2010). Transgender youth also may face increased discrimination throughout their transition process (e.g., as they begin to express their affirmed gender in the way they dress, make legal changes to their name and sex designation, undergo hormone therapy or medical procedures). These additional forms of resistance and discrimination are likely to act as barriers to learning and academic performance through many of the pathways we have described. For instance, school resistance and lack of safety may lead to increased absenteeism, diminished health, heightened vigilance, and isolation among transgender youth (Garofalo et al., 2006; Grossman & D'Augelli, 2006; Rosenberg, 2002), all of which may impede learning and academic progress. It is important to highlight these needs because few schools receive adequate training for working with transgender youth. Although some schools may receive training and service learning opportunities that focus on lesbian, gay, and bisexual issues, these trainings rarely provide sufficient time to review and discuss the specific needs of transgender youth and ways to meet their needs. In essence, the lack of attention to transgender youth is a major limitation in the field that extends not only to issues of academic disparities but to disparities in general. As such, greater inclusion of transgender youth is a necessity in order for research on sexual minority and gender-variant youth to make continued advances.

6. PROGRAMMING AND POLICY

6.1. Promoting Academic Resilience in the Face of Victimization

As research continues to identify the various pathways by which victimization disrupts and impedes learning and academic performance among sexual

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minority and gender-variant youth, it is equally important to identify factors that attenuate and buffer against such negative effects. There has been a growing interest in understanding resilience among sexual minority and gender-variant youth and factors that promote their healthy development. Positive youth development models provide a framework for this expansion (Damon, 2004; Larson, 2000). These models highlight that youth possess inner strengths that interact with assets in their environments to promote healthy development. In addition, these models point to various approaches that facilitate this process such as capitalizing on youth strengths, empowering youth, placing them in leadership positions, and providing opportunities to develop skills. These models also point to the need for supportive adults throughout this process. Intellectual development and academic excellence is a major domain of positive youth development, yet it has been less studied than psychological health among sexual minority and gender-variant youth. Using the positive youth development framework, we consider several relevant factors at the individual and systems level that may attenuate the effects of victimization on academic outcomes and that may promote resilience among sexual minority and gender-variant youth.

6.2. Peer, Parent, and General Adult Support

Much research on buffers against the negative effects of victimization has focused on the effects of peer, parent, or general adult support. Although sexual minority and gender-variant youth often have less access to support from these individuals due to discrimination or rejection (D'Augelli, Hershberger, & Pilkington, 1998; Ryan et al., 2009), the support that they do receive can be crucial in fostering their resilience. This has been considered primarily in relation to mental health outcomes, yet these same sources (e.g., peers, parents, or other adults) also can provide support relevant for academic achievement and career development.

There are several ways in which support operates to promote resilience. There has been consistent evidence that peer, parent, and adult support are directly associated with psychological well-being for sexual minority and gender-variant youth (Doty, Willoughby, Lindahl, & Malik, 2010; Eisenberg & Resnick, 2006; Hershberger & D'Augelli, 1995; Murdock & Bolch, 2005). These direct associations suggest that social support promotes overall health and well-being among these youth. In contrast, there has been mixed evidence for whether support from these sources attenuates the effects specifically tied to victimization. The attenuating effects have been weak or nonsignificant when considering general support from parents or peers

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(Murdock & Bolch, 2005; Poteat et al., 2011), but they have been significant when considering sexual orientation-specific support (Doty et al., 2010). There are several reasons why parent support may be an inconsistent buffer of victimization effects among sexual minority youth. Many of these youth face or fear parental rejection of their sexual orientation (D'Augelli et al., 1998; Ryan et al., 2009). Therefore, they may hesitate to approach their parents for support when they are victimized, particularly if the victimization is homophobic and they are not yet out to their parents. This same concern and hesitance may apply to accessing support from peers, teachers, and other adults. Youth who are not out to these individuals may fear inadvertent disclosure or discovery of their sexual orientation in the process of help seeking. This may be perceived as an additional stressor when youth are unsure of whether these individuals would be affirming of their sexual orientation. This issue might suggest why sexual orientation-specific support has more consistent moderating effects. This specific kind of support conveys affirmation and may lead youth to feel safer in their help seeking behavior from these individuals.

We raise several issues that warrant further consideration in how peer, parent, and adult support are tied to academic outcomes for sexual minority and gender-variant youth. Notably, most of the research on how support promotes resilience among these youth has considered its moderating effects on the association between victimization and mental health outcomes. Indeed, it is likely that the primary intent of the support that sexual minority and gender-variant youth seek and receive is meant to ensure their safety and promote their psychological health, as this is a fundamental need for these youth. To what extent should we expect support to moderate the effects of victimization on academic outcomes? Most studies in this area have assessed support in terms of emotional and social support (e.g., caring, acceptance, warmth, encouragement), which are closely tied to mental health. As we have noted, diminished mental health mediates the association between victimization and academic outcomes (Juvonen et al., 2000). Therefore, social support may promote academic resilience in an indirect manner. Specifically, it may do so by attenuating the effects of victimization on mental health concerns, which in turn affect academic outcomes (Poteat et al., 2011). As such, the effects of victimization on academic performance through diminished mental health may apply more strongly for sexual minority and gender-variant youth who receive less social support. At the same time, other forms of support (e.g., instrumental or informational support such as tutoring) may have direct ties to academic performance for

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sexual minority and gender-variant youth. These forms of support have been understudied and should be considered in future research.

Our limited knowledge of how to promote academic excellence among sexual minority and gender-variant youth underscores the need for researchers to consider more expansive resilience-based models that incorporate academic outcomes. For example, public single-sex schooling has become more common in the U.S. in response to proponents claims that such settings reduce gender-related bullying and sexual harassment, and thereby promote achievement (see Bigler, Hayes, & Liben, 2014 [Chapter 7 of this volume]). Some critics are skeptical that such settings are associated with positive academic outcomes among sexual minorities and gender-variant youth, yet little empirical work on the topic exits. This call for attention to academic outcomes is not to detract attention from the serious and immediate safety needs of sexual minority and gender-variant youth in schools or to suggest a diversion of resources. Rather, it is meant to add to this scope of work to ensure that, in the face of such adversity, sexual minority and gender-variant youth also receive adequate support to meet their academic needs and to work toward their long-term career aspirations. These goals are crucial for promoting a more comprehensive sense of long-term well-being and resilience for these youth.

6.3. The Role of Gay–Straight Alliances

Structures at the systems level also may promote academic resilience among sexual minority and gender-variant youth. Some work has focused on the effects of Gay–Straight Alliances (GSAs). GSAs have a strong foundation in positive youth development models. They are meant to be youth-led and adult-supported groups that provide a setting for gender-variant, sexual minority, and heterosexual youth to receive emotional and social support, to socialize, and to lead advocacy efforts that address discrimination and inequality in the school and community (Griffin, Lee, Waugh, & Beyer, 2004; Russell, Muraco, Subramaniam, & Laub, 2009). GSAs have a strong potential to promote resilience among a large number of youth because they are directly situated within schools and are showing expansive growth across the country (GLSEN, 2012; GSA Network, 2013).

The presence of GSAs in schools has been linked to a number of academic and health-related indices. For example, sexual minority youth in schools with GSAs report more welcoming climates for diversity, greater perceived safety, less truancy, lower suicidality, less frequent engagement

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in risky sexual behavior, and lower substance use than those in schools without GSAs (Goodenow et al., 2006; Poteat, Sinclair, DiGiovanni, Koenig, & Russell, 2013; Szalacha, 2003; Walls, Kane, & Wisneski, 2010). It appears that GSAs have the potential to promote resilience across a broad range of factors. Researchers have referred to the dual purpose of GSAs to provide support to their members and to address broader systems of inequality to explain these patterns.

These GSA-related findings are encouraging, yet several issues should be considered in relation to how GSAs promote resilience in academic domains. As stated, one of the primary intentions of GSAs is to provide emotional and social support to members. This is distinct from the provision of academic support. In fact, most of the research on GSA-related effects has focused on indices of psychological and behavioral health and social well-being. Fewer studies have examined purely academic outcomes in relation to GSA membership, and their findings have been mixed. For example, while some studies have found that students who attend schools with GSAs report higher GPAs (Walls et al., 2010), the magnitude of this difference was not large. Other studies have failed to identify GPA differences based on GSA presence when controlling for other school characteristics (e.g., school size or socioeconomic status; Poteat, Sinclair, et al., 2013). Although it is powerful to see that GSAs may promote resilience, it would be unfair and unreasonable to expect GSAs to serve as an all-encompassing panacea in promoting resilience across all domains of youth development. They should not be expected to serve as the sole source or primary mechanism of change for either psychosocial or academic resilience (Poteat, Sinclair, et al., 2013). Rather, it would be optimal and perhaps necessary for GSAs to operate in conjunction with other programs and groups within the school (e.g., guidance offices, administrators and teachers, other student groups) to address large-scale concerns such as prejudice and discrimination.

Although the primary emphasis of GSAs may not be to provide direct academic support or enhancement to their members, GSAs are nonetheless in a position to advocate for programs, policies, or other efforts that could improve the educational experience for sexual minority and gender-variant students. This could be considered as part of the advocacy-based nature of GSAs. For instance, GSAs may petition for policies and programs such as inclusive curricula (e.g., course-based material that is representative of sexual minority and gender-variant individuals). We later discuss how these materials could enrich the learning process and academic performance for sexual minority and gender-variant youth. Because many GSAs seek to engage in

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advocacy efforts that benefit the entire school and not only immediate club members (Griffin et al., 2004), these actions could result in positive outcomes for the general student population. Thus, even sexual minority and gender-variant youth who are not members of GSAs may derive benefits from their actions.

6.4. Broader Extracurricular Opportunities

More broadly than GSAs, many youth participate in a range of school-based extracurricular activities and groups (e.g., sports or clubs) as a part of their educational experience. These settings also can promote resilience and healthy youth development (Fredricks, 2012). Participating in extracurricular activities is associated with higher GPAs and school completion rates (Darling, 2005; Eccles & Barber, 1999; Feldman & Matjasko, 2005) and higher self-esteem (Blomfield & Barber, 2009). Extracurricular participation also has been linked to greater interpersonal competence, school engagement, sense of belonging, and educational aspirations (Denault & Poulin, 2009; Marsh & Kleitman, 2002).

There are several explanations for the academic benefits derived through extracurricular involvement. Structured activities promote physical safety, supportive relationships, positive social norms, and self-efficacy (Eccles & Gootman, 2002). Participation in extracurricular activities increases school connectedness, self-worth, and sense of school belonging (Libbey, 2004). In essence, extracurricular sports and clubs are positioned to promote healthy development in many of the same domains around which sexual minority and gender-variant youth experience greater hardships due to victimization and marginalization in schools.

Despite the benefits of extracurricular involvement, discrimination is evident in these contexts for many sexual minority and gender-variant youth. Organized sports can be particularly hostile environments in which hegemonic masculinity and sexual prejudice are pervasive (Gill, Morrow, Collins, Lucey, & Schultz, 2010). These unwelcoming and unsafe climates may discourage or explicitly prevent sexual minority and gender-variant youth from participating in these activities (Gill et al., 2010; Mishna, Newman, Daley, & Solomon, 2009). Ultimately, social exclusion and homophobic climates within certain extracurricular groups deprive many sexual minority and gender-variant youth of the benefits of participating in them.

Males who are involved in sports are socialized to conform to a restricted masculine and heterosexual identity; those who deviate from these norms

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often face ridicule, harassment, or physical violence (Gill et al., 2010; Messner, 1992). Furthermore, males who wish to participate in stereotypically female-oriented activities, such as cheerleading, often are victimized based on their perceived sexual orientation (Barron & Bradford, 2007). Sexual minority males who are involved in sports often feel pressured to conceal their sexual orientation out of fear of reprisal or rejection (Griffin, 1993). Although sexual minority males involved in sports may avoid being targets of discrimination because peers are less likely to suspect them to be gay or bisexual, this carries a substantial psychological and social cost of hiding their identity and facing exposure to prejudice and homophobic behavior from peers and adult figures (Griffin, 1993).

Sexual minority females also face barriers and challenges to sports participation, although they differ in some ways from those faced by males. (Barriers to physical activity and sport participation faced by girls in K-12 education are reviewed by Solmon, 2014 [Chapter 4 of this volume].) For instance, males who participate in sports are validated for their heightened displays of masculinity, yet females often feel pressured to compensate with heightened femininity to offset their engagement in stereotypically masculine behavior (Griffin, 1993; Stoelting, 2011). Consequently, whereas males who participate in sports may be less likely to be considered gay or bisexual, females who participate in sports may be more likely to be considered lesbian or bisexual (Gill et al., 2010). At the same time, it is unclear whether homophobic expressions and pressure to conform to feminine norms arise from within the setting (e.g., from teammates or coaches) as they do for males, or whether they originate from outside the setting (e.g., from other students within the school). In general, little research has explored the experiences of sexual minority youth athletes or whether experiences differ for male and female youth athletes due to gender differences in sexual prejudice and its intersection with traditional gender norms. Nevertheless, attention to these issues would be quite valuable for efforts to develop tailored programming to counter sexual prejudice and sexism in these settings within which sexual minority youth may face the greatest hostility.

Adding to these concerns, some school policies perpetuate the marginalization of transgender youth. Participation in some clubs, particularly sports teams, is sometimes restricted by one's biological sex (Barron & Bradford, 2007). For example, transgender females may not be allowed to participate in women's sports teams. As with most other areas of research, however, there has been a severe lack of attention to the experiences of transgender youth in these contexts. As schools begin to address issues related to gender identity and

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expression, however, there has been a growing awareness that this constitutes a priority area within research, practice, and policy work.

Although sexual minority and gender-variant youth face barriers and resistance to their involvement in sports and clubs, recent research suggests that these outlets may promote overall well-being and positive academic outcomes for those who are involved in them (Toomey & Russell, 2013a). If barriers to extracurricular involvement can be diminished, these settings could be a major source of support and may promote healthy social and academic development for sexual minority and gender-variant youth. Thus, researchers should give much greater attention to how school-based extracurricular involvement may foster social relationships, improve mental health, and promote academic achievement among sexual minority and gender-variant youth.

6.5. School-Wide Policies and Academic Programs

We have used the school context as a setting for our discussion of various individual psychological and proximal social factors that contribute to variability in the learning experiences and academic outcomes of sexual minority and gender-variant youth. In addition, it is important to consider even broader school characteristics that have the potential to affect the academic outcomes of these youth. There has been growing attention to variability among sexual minority youth as a function of the policies and programs that exist at their schools (Chesir-Teran & Hughes, 2009; Hatzenbuehler & Keyes, 2013; O'Shaughnessy, Russell, Heck, Calhoun, & Laub, 2004; Szalacha, 2003). In this section, we extend our focus to this systems level and suggest how several school-wide policy and programming efforts may promote resilience among these youth.

6.6. Anti-discrimination Policies

Passage of enumerated anti-bullying and anti-discrimination school policies is a prominent issue that continues to undergo contentious debate in many states and at the federal level. Although nearly all states and the District of Columbia have passed some form of anti-bullying legislation, at present only a small percentage of these states have passed enumerated policies that extend explicit protection to students based on their actual or perceived sexual orientation or gender identity and expression (Russell, Kosciw, Horn, & Saewyc, 2010). Arguments in support of these policies have emphasized the need to ensure safe environments for students from social groups that

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historically have faced disproportionately higher rates of victimization (Jacob, 2013; Russell et al., 2010). In this sense, enumerated policies are not limited to sexual orientation and gender identity or expression; many include other protected categories such as disability, race, religion, or sex. Yet, sexual orientation and gender identity and expression often are the most contested categories of those included in these policies (Horn, Szalacha, & Drill, 2008; Russell et al., 2010).

Although there has been marked resistance to these inclusive policies, emerging empirical findings suggest that such policies contribute to greater safety for youth in these schools. In the state of Oregon, lesbian and gay high school students in counties where a larger number of districts had adopted enumerated policies reported fewer suicide attempts than those in counties where fewer districts had adopted these policies (Hatzenbuehler & Keyes, 2013). Data from students across the state of California also suggest that sexual minority youth who attend schools with enumerated protective policies report greater perceptions of safety than youth in schools without these policies (O'Shaughnessy et al., 2004). Finally, reports from the Gay, Lesbian, and Straight Education Network (GLSEN) that draw from their school climate survey of youth also suggest that sexual minority and gender-variant students in states with enumerated policies report hearing fewer homophobic epithets at school and observe more consistent staff intervention during these instances (Kosciw et al., 2008).

Although these policies appear to serve their intended purpose, sexual minority and gender-variant youth in schools with these policies continue to face some level of discrimination. Researchers and policies advocates have thus pointed to issues in how these policies are applied. For instance, there is a need to address how these policies are implemented, the extent to which they are consistently enforced, and the extent to which students understand the procedures for reporting discrimination (Hansen, 2007; Russell et al., 2010).

As researchers consider issues related to policy implementation, they also must consider the underlying mechanisms by which these policies promote safety and well-being for sexual minority and gender-variant youth. Some scholars have proposed that enumerated policies provide a clear indication to students and adults that sexual minority and gender-variant youth must be protected (Russell et al., 2010). This formal recognition may empower youth to report instances of discrimination. It may also encourage adults to be more responsive to these instances, addressing concerns from students who have reported that teachers often do not intervene during instances of

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homophobic behavior (Elze, 2003). Such policies may also lead teachers to perceive and experience administrative support for their efforts to challenge sexist and homophobic behavior in their classrooms. Empirical research has not yet tested these mechanisms.

It is also important to consider the effects of these protective policies on academic outcomes and to identify how these policies operate to promote learning and academic success. Although these policies exist at the broader systems level, they may exert a significant influence on individual students' experiences that are relevant to learning and academic performance. For instance, if these policies prompt adults to intervene more consistently when homophobic behavior occurs, this may eventually lead to more welcoming school climates and to less frequent victimization of sexual minority and gender-variant youth. In turn, this may lead these youth to attend school more consistently due to lowered safety concerns, and they may experience less vigilance, less engagement in harmful externalizing coping strategies, and improved psychological health. Also, youth may feel more welcomed to approach teachers for emotional and academic support. Each of these outcomes ultimately may create more optimal conditions for learning, opportunities to take part in extracurricular activities for further enrichment, and the necessary support to excel academically.

6.7. Inclusive Curriculum

Course material and discussions on sexual orientation-related diversity issues are limited or absent in many classrooms (Lipkin, 1999; Rogers & Mosley, 2006; Wills, 2001). As a complement to enumerated nondiscrimination policies, an inclusive curriculum has the potential to promote positive learning experiences for sexual minority and gender-variant youth. In this case, an inclusive curriculum would represent the integration of sexual orientation and gender diversity topics and the representation of sexual minority and transgender individuals within the standard curriculum. As examples, this could involve the inclusion of the sexual minority civil rights movement in history courses, acknowledging that certain pioneers in many fields (e.g., the arts and sciences) identified as lesbian, gay, bisexual, or transgender, or ensuring that word problems in math assignments are not strictly heteronormative (e.g., only referring to opposite-sex couples or "nuclear" families).

There are a number of sources that provide inclusive lesson plans appropriate for different grade levels and course subjects (e.g., GLSEN; www.glsen.org). In addition, extensive materials are available to supplement

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the standard curriculum (e.g., Welcoming Schools from the Human Rights Campaign, www.welcomingschools.org; films with accompanying curricula on family diversity, gender identity and expression, and bias-based harassment from Groundspark, www.groundspark.org). In this section, we review some of the initial research related to inclusive curricula and suggest various pathways by which inclusive curricula could promote positive developmental outcomes, including those related to academic achievement.

There are some limited findings that an inclusive school curriculum is associated with greater perceptions of safety among students in those schools. One report found that students who attended schools that included sexual minority issues in the curriculum reported greater perceptions of safety than students in schools where this material was absent (Russell, Kostroski, McGuire, Laub, & Manke, 2006). Similarly, students in schools with an inclusive curriculum report greater perceptions of positive diversity climates at their schools (Szalacha, 2003). The data from GLSEN suggest a similar pattern; students who reported learning about sexual minority issues in their schools reported less victimization and greater perceptions of safety than their peers who did not report such learning opportunities (Kosciw et al., 2008). These findings point to the value and importance of an inclusive curriculum as it relates to the fundamental safety concerns of youth. Given that heterosexual and sexual minority youth can experience homophobic victimization and victimization for gender-variant behavior (Pascoe, 2007; Phoenix et al., 2003; Plummer, 2001; Poteat & Espelage, 2007), an inclusive curriculum that promotes respect for these forms of diversity stands to benefit all youth along indices of psychological and social well-being. Nevertheless, there is a dearth of research on whether such curricula have effects on academic outcomes.

Before we proceed to outline the various pathways through which inclusive curricula may promote academic performance, we would first argue that students' knowledge of diverse groups is itself a necessary and legitimate area of academic competence. Students must be aware of how their own attitudes, beliefs, and behaviors are situated within a diverse society; they must have knowledge and an understanding of the diverse views and experiences of others; and they must be able to relate and work successfully with their peers in increasingly diverse schools and workforces (Gurin, Dey, Hurtado, & Gurin, 2002). In this way, an inclusive curriculum is directly connected to meeting these needs and this facet of academic competence.

We suspect that an inclusive curriculum promotes positive learning experiences and greater overall academic performance for sexual minority and gender-variant youth through several pathways. First, an inclusive

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curriculum serves to destigmatize sexual minority and gender-variant youth. It also presents opportunities for learning and dialogue about stereotypes and about the experiences of sexual minority and gender-variant individuals. Destigmatization efforts carry important implications for all students. Sexual minority youth experience invisibility within heteronormative school systems (Black, Fedewa, & Gonzalez, 2012; Chesir-Teran & Hughes, 2009). Thus, it is imperative that they begin to find themselves equally represented and valued. In addition, it is important for heterosexual youth to learn about sexual minorities as a diverse group in society. Furthermore, it is important for all youth to see that their schools recognize and value sexual minority and gender-variant youth as a part of the school community.

Research on intergroup contact also may help to explain how an inclusive curriculum leads to decreased bias against sexual minority and gender-variant individuals. There is strong support for the association between direct contact with sexual minorities (e.g., through friendships or family connections) and lower sexual prejudice (Heinze & Horn, 2009; Smith, Axelton, & Saucier, 2009). (Empirical support for a parallel association between direct contact with other-gender peers and lower gender prejudice is reviewed by Martin, Fabes, and Hanish, 2014 [Chapter 5 of this volume].) In addition to direct contact, researchers have given attention to the influence of extended contact (Crisp & Turner, 2009). Extended contact can range from knowing that another friend has sexual minority or gender-variant friends to simply imagining interactions with a sexual minority or gender-variant individual. The material and discussions that are embedded within inclusive curricula may provide opportunities for these types of extended contact. For instance, even in the absence of "out" sexual minority youth in the classroom, students may learn that another classmate has sexual minority friends, or students may begin to consider what it would be like to interact with sexual minority peers.

Inclusive curricula also may promote safer and more welcoming schools for sexual minority and gender-variant youth. Some youth may come to reconsider their language and actions that contribute to hostile climates for sexual minority and gender-variant youth and this may lead to decreased discriminatory behavior against these youth. Additionally, as sexual minority and gender-variant youth begin to see themselves represented and acknowledged by teachers and administrators in respectful ways, they may feel a greater sense of support and school belonging. In effect, these processes may partly explain the extant findings that sexual minority youth in schools with inclusive curricula report greater safety and positive climates (Kosciw et al., 2008; Russell et al., 2006; Szalacha, 2003).

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Through destigmatization, direct or extended contact opportunities, and efforts to promote more respectful and welcoming climates, inclusive curricula ultimately may promote better academic outcomes for sexual minority and gender-variant youth. We expect this connection because under these circumstances, sexual minority and gender-variant youth are likely to face fewer affective, cognitive, and social stressors that impede many of the learning processes that we have discussed (e.g., high vigilance, concentration difficulties, school avoidance). Furthermore, because a number of heterosexual youth face homophobic victimization or victimization based on gender-variant behavior (Pascoe, 2007; Phoenix et al., 2003; Plummer, 2001; Poteat & Espelage, 2007), inclusive curricula also could benefit their ability to learn and perform at their maximum potential.

The effects of inclusive curricula on academic outcomes also may operate through pathways that directly represent individual learning processes. Sexual minority and gender-variant youth who see themselves represented in course material may connect with the material in more meaningful ways. Educators have noted the value of connecting with and relating to course material for students from other marginalized groups (Barbarin et al., 2014 [Chapter 10 of this volume]; Lee & Buxton, 2008; Matthews & Smith, 1994; Rowley et al., 2014 [Chapter 9 of this volume]). These youth also may feel a greater sense of self-efficacy in contributing to class discussions because they can more easily relate to the material. In addition, inclusive curricula across multiple course subjects may lead sexual minority and gender-variant youth to consider careers in fields that they may have initially considered inaccessible. In addition, sexual minority and gender-variant youth may feel more motivated to learn the material, may express greater interest in the subject, and may engage in the overall learning process more actively (e.g., through class participation, homework completion). All of these processes fit well within the SCCT model (Lent et al., 1994), in which learning experiences (e.g., exposure to inclusive curricula) predict self-efficacy and outcome expectations (e.g., confidence in contributing to discussions, anticipating good performance), which in turn predict interests and ensuing actions (e.g., studying, completing homework, reading more about the subject) that ultimately lead to performance outcomes (e.g., higher grades, working toward a career in that area). Thus, in addition to addressing broader social contextual concerns (e.g., improving intergroup relations, fostering more welcoming school climates), inclusive curricula may promote the

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academic performance of sexual minority and gender-variant youth through these more individual-based learning processes.

7. CONCLUSIONS

The discrimination that sexual minority and gender-variant youth experience affects many interrelated areas of their lives, ranging from their psychological, to social, to academic development. Consequently, sexual orientation-based and gender identity or expression-based disparities constitute major areas of research across many disciplines in the social sciences. In this chapter, we have focused on school and academic concerns faced by sexual minority and gender-variant youth, while considering how these concerns are connected to concerns in other domains. Our aim was to highlight the current state of research in this area and to stimulate a significant expansion of research by highlighting current limitations and issues that have not yet been addressed. In doing so, we have drawn from theories such as minority stress theory (Meyer, 2003), SCCT (Lent et al., 1994), and positive youth development (Damon, 2004) to propose several processes by which multiple factors could contribute to academic disparities, as well as processes that could promote academic resilience. These processes, as well as others, should be formally examined and tested in future research. In addition, there is a substantial need for research to evaluate the immediate and long-term effectiveness of prevention and intervention programs and the effectiveness of school policies in promoting the well-being of sexual minority and gender-variant youth. These evaluations should include attention to how such efforts enrich the learning experiences of these youth, promote their academic excellence, and provide opportunities for long-term career development.

These future directions represent an ambitious undertaking at the intersection of research, policy, and practice across multiple disciplines. This work necessitates a continued long-term commitment to address the array of factors at the individual and systems level that affect the learning and academic growth of sexual minority and gender-variant youth. Such work also requires awareness of the substantial diversity among these youth and attention to their unique needs and experiences. Ultimately, these advances stand to have a significant and long-lasting impact on the development of sexual minority and gender-variant youth well beyond their years of formal schooling and education.

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